

ZBM Email Automation System – Project Handover Guide

Executive Summary

This project is an automated email distribution system for Zone Business Managers (ZBMs) that processes sample dispatch data, generates hierarchical reports, and sends personalized emails with consolidated attachments. The system streamlines the reporting process for pharmaceutical sample distribution across a multi-tier sales organization (ZBM → ABM → TBM → HCP).

Project Overview

Purpose

Automate the monthly/periodic distribution of sample dispatch reports to Zone Business Managers, including:

- Consolidated sample dispatch data files
- Hierarchical summary reports showing ABM performance
- HTML-formatted email summaries

Business Context

The organization follows a hierarchical structure:

- **ZBM (Zone Business Manager)** - Top level, oversees multiple ABMs
- **ABM (Area Business Manager)** - Mid level, oversees multiple TBMs
- **TBM (Territory Business Manager)** - Field level, interacts with HCPs
- **HCP (Healthcare Professional)** - End recipient of sample dispatches

System Architecture

Workflow Overview

...

1. Data Input → 2. Generate Consolidated Files → 3. Generate Reports → 4. Send Emails

...

Processing Flow

1. **Input**: Sample Master Tracker Excel file with dispatch records
2. **Consolidation**: Creates individual Excel files per ZBM with detailed dispatch data
3. **Reporting**: Generates hierarchical summary reports showing ABM-level metrics

4. **Distribution**: Automated email creation with attachments via Outlook

Code Files Description

1. `create\_zbm\_consolidated\_files (4).py`

Purpose: Creates detailed consolidated Excel files for each ZBM

Key Functions:

- Reads Sample Master Tracker data
- Applies business rules from `logic.xlsx` to compute final status
- Deduplicates data to prevent double-counting
- Creates one consolidated file per ZBM with columns:
 - Request IDs, Customer codes, SKU details
 - Invoice/Dispatch/Delivery dates
 - Status information (Request Status, Final Status)
 - RTO (Return to Origin) reasons
 - Hierarchy information (ABM, TBM details)

Output:

- Folder: `ZBM\_Consolidated\_Files\_[timestamp]`
- Files: `ZBM\_Consolidated\_[ZBM\_Code]\_[ZBM\_Name]\_[date].xlsx`

Processing Logic:

- Filters and cleans data (removes null ZBM/ABM codes)
- Computes "Final Status" using rule mapping from `logic.xlsx`
- Formats date columns (removes timestamps)
- Sorts by ABM Territory Code and Request ID

2. `create\_zbm\_hierarchical\_reports (9).py`

Purpose: Creates summary reports showing ABM-level performance metrics

Key Functions:

- Aggregates data at ABM level under each ZBM
- Calculates key performance metrics:
 - **Section A**: Request Cancelled (Out of Stock, On Hold, Not Permitted)
 - **Section B**: Action Pending at HO (Headquarters)
 - **Section C**: Sent to HUB (sum of D+E+F)

- **Section D**: Pending for Invoicing
- **Section E**: Pending for Dispatch
- **Section F**: Requests Dispatched (sum of G+H+I)
- **Section G**: Delivered
- **Section H**: Dispatched & In Transit
- **Section I**: RTO (Return to Origin) with breakdown:
 - Incomplete Address
 - Doctor Non-Contactable
 - Doctor Refused to Accept
 - Hold Delivery

Output:

- Folder: `ZBM\_Reports\_[timestamp]`
- Files: `ZBM\_Summary\_[ZBM\_Code]\_[ZBM\_Name]\_[date].xlsx`

Report Structure:

- Uses template file: `zbm\_summary.xlsx`
- Shows one row per ABM with metrics
- Includes total row at bottom
- Preserves template formatting (colors, borders, fonts)

Validation:

- Verifies tally: $A + B + C = \text{Total Requests}$
- Checks that $C = D + E + F$
- Ensures $F = G + H + I$
- Validates RTO breakdown matches RTO total

3. `send\_zbm\_emails (3).py`

Purpose: Automates email creation and distribution to ZBMs

Key Functions:

- Locates generated consolidated files and reports
- Reads ZBM contact information from Sample Master Tracker
- Extracts summary table from Excel report and converts to HTML
- Creates Outlook emails with:
 - HTML body showing summary metrics
 - Consolidated file as attachment
 - Optional CC to ABMs
 - Sent on behalf of: EPD_SFA@abbott.com

****Email Components**:**

- ****Template****: `email\_template\_ZBM.html` (Jinja2 format)
- ****Subject****: "Sample Direct Dispatch - ZBM Summary Report as of [date]"
- ****Body****: Personalized with ZBM name and HTML summary table
- ****Attachment****: ZBM's consolidated Excel file

****Process**:**

1. Finds latest consolidated folder and reports folder
2. Reads ZBM list from Sample Master Tracker
3. For each ZBM:
 - Locates their consolidated file
 - Reads their summary report
 - Extracts data as HTML table (preserves Excel formatting)
 - Creates Outlook email (displays for review)
4. Logs all email activities

****Output**:**

- Folder: `ZBM\_Email\_Logs\_[date]`
- File: `email\_log.txt` with timestamp entries

Supporting Files

Required Input Files

1. ****Sample Master Tracker.xlsx**** (or similar name)
 - Main data source with all dispatch records
 - Must contain columns: ZBM Terr Code, ZBM Name, ZBM EMAIL_ID, ABM details, TBM details, Request Status, dates, etc.
2. ****logic.xlsx****
 - Contains business rules for status mapping
 - Sheet2: Maps combinations of statuses to "Final Answer"
 - Used to determine final status of each request
3. ****zbm_summary.xlsx****
 - Template file for summary reports
 - Contains pre-formatted structure with headers and styling
 - Must have "ZBM" sheet with proper column headers

4. ****email_template_ZBM.html****

- Jinja2 template for email body
- Variables: `zbm\_name`, `zbm\_code`, `current\_date`, `summary\_table`

Execution Instructions

Prerequisites

- ****Python 3.x**** installed
- ****Required Libraries****:
...
pandas
openpyxl
jinja2
pywin32 (for Outlook automation)
...
- ****Microsoft Outlook**** installed and configured
- All input files in the same directory as scripts

Installation

```
```bash
pip install pandas openpyxl jinja2 pywin32
```
```

Execution Sequence

****Step 1: Generate Consolidated Files****

```
```bash
python create_zbm_consolidated_files.py
```
```

- Creates: `ZBM\_Consolidated\_Files\_[timestamp]` folder
- Time: ~2-5 minutes depending on data size

****Step 2: Generate Summary Reports****

```
```bash
python create_zbm_hierarchical_reports.py
```
```

- Creates: `ZBM\_Reports\_[timestamp]` folder
- Time: ~2-5 minutes depending on data size

****Step 3: Send Emails****

```
```bash
python send_zbm_emails.py
```
```

- Creates: `ZBM\_Email\_Logs\_[date]` folder
- Displays Outlook emails for review (not auto-sent)
- User must manually send each email

Important Notes

- Scripts must be run in sequence (1 → 2 → 3)
- Each script validates required folders/files exist
- Email script displays emails for review; user sends manually
- Always verify output folders were created successfully

Data Processing Logic

Final Status Calculation

The system computes a "Final Answer" for each request by:

1. Grouping all statuses for a unique Request ID
2. Normalizing status text (lowercase, trimmed)
3. Creating a sorted tuple of unique statuses
4. Looking up the combination in `logic.xlsx` Sheet2
5. Assigning the mapped "Final Answer"

****Example**:**

- Request statuses: ["Dispatched", "Delivered"]
- Lookup key: ('delivered', 'dispatched')
- Final Answer: "Delivered"

Deduplication Strategy

To ensure accurate counts:

- ****Consolidation****: Deduplicated at (Request ID + ZBM + ABM) level
- ****Reporting****: Uses deduplicated data to count unique requests
- ****Rationale****: Same request may appear in multiple rows for different line items/SKUs

RTO Reason Prioritization

When a request has multiple RTO reasons, uses priority:

1. Incomplete Address (highest priority)

2. Doctor Refused to Accept
3. Doctor Non-Contactable
4. Hold Delivery (lowest priority)

Each unique Request ID counted in only ONE category.

Validation & Quality Checks

Built-in Validations

1. **Tally Validation**

- Verifies: Requests Raised = $A + B + C$
- Verifies: Sent to HUB (C) = $D + E + F$
- Verifies: Requests Dispatched (F) = $G + H + I$
- Prints warnings for mismatches

2. **RTO Breakdown Validation**

- Ensures: RTO Total = Sum of all RTO reason categories
- Prints mismatches with details

3. **File Existence Checks**

- Email script verifies consolidated files and reports exist
- Warns if files not found for specific ZBMs

4. **Data Quality Checks**

- Removes records with null/empty ZBM or ABM codes
- Filters out invalid email addresses ('0', '0.0')
- Validates required columns exist in input files

Debug Output

Scripts print comprehensive logs:

- Record counts at each processing stage
- Unique ZBM/ABM counts
- Sample data rows for verification
- Warnings for any data quality issues
- Success/failure status for each operation

Common Issues & Troubleshooting

Issue 1: "Could not find ZBM_Consolidated_Files folder"

****Cause**:** Step 1 not run or failed

****Solution**:** Run `create\_zbm\_consolidated\_files.py` first

Issue 2: "Could not find ZBM_Reports folder"

****Cause**:** Step 2 not run or failed

****Solution**:** Run `create\_zbm\_hierarchical\_reports.py` second

Issue 3: Tally mismatches in reports

****Cause**:** Unmapped statuses in logic.xlsx

****Solution**:**

- Check console output for "unmapped requests"
- Add missing status combinations to logic.xlsx Sheet2
- Re-run processing scripts

Issue 4: Email attachments fail to attach

****Cause**:** File path issues or file doesn't exist

****Solution**:**

- Verify consolidated file exists in output folder
- Check console for full file path
- Ensure no special characters in ZBM name

Issue 5: Missing columns error

****Cause**:** Input file column names changed

****Solution**:**

- Verify Sample Master Tracker has all required columns
- Update column names in script if necessary

Issue 6: Outlook automation not working

****Cause**:** Outlook not running or permission issues

****Solution**:**

- Start Outlook application
- Check Windows firewall/antivirus settings
- Run Python as administrator if needed

Maintenance & Updates

Regular Maintenance Tasks

1. **Monthly** (Before each run):
 - Verify Sample Master Tracker is updated with latest data
 - Check logic.xlsx has current status mappings
 - Test with small dataset first
2. **Quarterly**:
 - Review and update email template if needed
 - Archive old output folders
 - Verify library versions are current
3. **As Needed**:
 - Add new status mappings to logic.xlsx
 - Update ZBM hierarchy if organization changes
 - Modify report template for new metrics

How to Modify

Add New Column to Consolidated File:

1. Add column name to `required\_columns` list (line 27)
2. Add column to `consolidated\_columns` list (line 149)

Add New Metric to Summary Report:

1. Calculate metric in ABM loop (around line 200)
2. Add to `summary\_data.append()` dictionary (line 280)
3. Update `zbm\_summary.xlsx` template with new column
4. Add column mapping in `create\_zbm\_excel\_report()` (line 400)

Change Email Template:

1. Edit `email\_template\_ZBM.html`
2. Use Jinja2 syntax for variables: `{ { variable\_name } }`
3. Test with one ZBM before full run

Update Status Rules:

1. Open `logic.xlsx` Sheet2
2. Add row with status combination and Final Answer
3. Re-run processing scripts

Performance Considerations

Expected Processing Times

- **Consolidated Files**: 2-5 minutes for ~50 ZBMs
- **Summary Reports**: 2-5 minutes for ~50 ZBMs
- **Email Creation**: 1-2 seconds per email (manual send)

Resource Usage

- **Memory**: ~500MB-1GB for typical dataset
- **Disk Space**:
 - Consolidated files: ~100-500KB per ZBM
 - Summary reports: ~50-100KB per ZBM
 - Total: ~50-100MB per run

Optimization Tips

- Filter data early to reduce processing
- Use `.copy()` when creating subsets to avoid warnings
- Close Excel files after processing
- Archive old output folders regularly

Security & Compliance

Data Handling

- **PII**: Contains doctor names, contact info - handle per privacy policy
- **Email Addresses**: ZBM, ABM, TBM emails - verify accuracy
- **Business Data**: Sample dispatch volumes - confidential

Best Practices

- Run scripts on secure, company-approved systems
- Don't share output files via unsecured channels
- Delete test outputs containing real data
- Review emails before sending
- Keep audit logs (email_log.txt)

Compliance Notes

- Email sent on behalf of: EPD_SFA@abbott.com
- Ensure sender delegation is authorized
- Retain logs per company retention policy
- Follow data protection regulations

Contact & Support

For Questions About:

- **Business Logic**: Contact ZBM reporting team
- **Data Issues**: Contact data management team
- **Technical Issues**: Contact IT support or original developer
- **Script Modifications**: Refer to this document and code comments

Documentation

- Code comments in each Python file
- This handover document
- Template files (zbm_summary.xlsx, email_template_ZBM.html)

Version History

| Version | Date | Changes | Author |
|---------|-----------|------------------------|----------------------|
| ----- | ----- | ----- | ----- |
| 1.0 | [Current] | Initial implementation | [Original Developer] |

Appendix

Key Terminology

- **ZBM**: Zone Business Manager
- **ABM**: Area Business Manager
- **TBM**: Territory Business Manager
- **HCP**: Healthcare Professional
- **HO**: Head Office
- **RTO**: Return to Origin
- **SKU**: Stock Keeping Unit

File Naming Conventions

- Consolidated: `ZBM\_Consolidated\_[Code]\_[Name]\_[Date].xlsx`
- Reports: `ZBM\_Summary\_[Code]\_[Name]\_[Date].xlsx`
- Folders: `[Type]\_[Timestamp]`

Sample Command Sequence

```
``bash
```

Complete workflow

```
python create_zbm_consolidated_files.py
```

```
python create_zbm_hierarchical_reports.py
```

```
python send_zbm_emails.py
```

Review outputs in created folders

Manually send emails from Outlook

```
...
```

```
---
```

****Document End****

For additional assistance, please refer to inline code comments or contact the technical team.